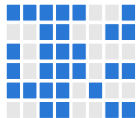
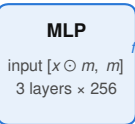


### (a) Pre-training, once per dataset

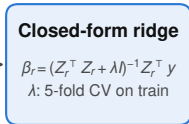
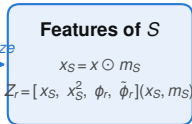
rows  $\times$  all columns held,  
random visibility mask  $m$



■ visible ■ hidden



freeze



#### Objective

predict targets + reconstruct hidden columns; freeze last hidden layer  $\phi_r$ ; untrained twin  $\tilde{\phi}_r$  (seeds  $r=1, 2, 3$ )

### (b) Query ( $S, y$ ): subset-specific readout

#### Estimate

$\hat{y} = \frac{1}{3} \sum_r Z_r \beta_r$ ;  $\hat{V}_y(S) = R_{\text{test}}^2(\hat{y}) + \text{closure}$   
safeguards: s.d. floor; clip features,  $\hat{y}$  to train range

### (c) Scan–certify

#### Scan

$\hat{V}_y$  for all  $|S| \leq K+1 \rightarrow$   
 $\hat{M}^{(K)} + \text{ranked backgrounds}$

#### Certify

retrain attackers only on the top- $k$  backgrounds  $\rightarrow L^{(K)} \leq M^{(K)}$